

演算法概論作業二

Spring 2026

作業二要做的事

- 判斷一個簡單無向圖是否「連通且有至少兩個cycles」，如果是，再判斷該圖是否有至少三個cycles

輸入的格式

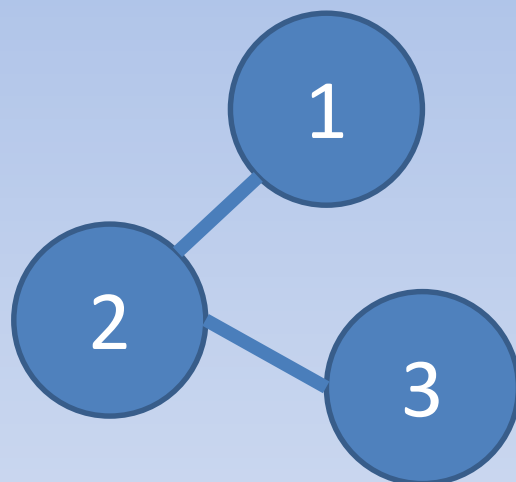
- 第一行是一個正整數 $n \leq 9$ ，用以表示簡單無向圖 $G = (\{1, 2, \dots, n\}, E)$ 的點數
- 之後的 $|E|$ 行表示所有 G 的邊，其中每一行恰含有一個邊之兩端點，以空白隔開
- 最後一行寫的是-1
- 從standard input給輸入

輸出的格式

- 如果 G 是連通的且 G 擁有至少兩個cycles，則輸出“y”後換行，否則請輸出“n”後換行
- 接著，如果 G 連通且擁有至少三個cycles，再輸出“There are at least three cycles.”

Sample input 1

3
1 2
3 2
-1



Sample output 1

n

Sample input 2

6

1 2

4 2

1 3

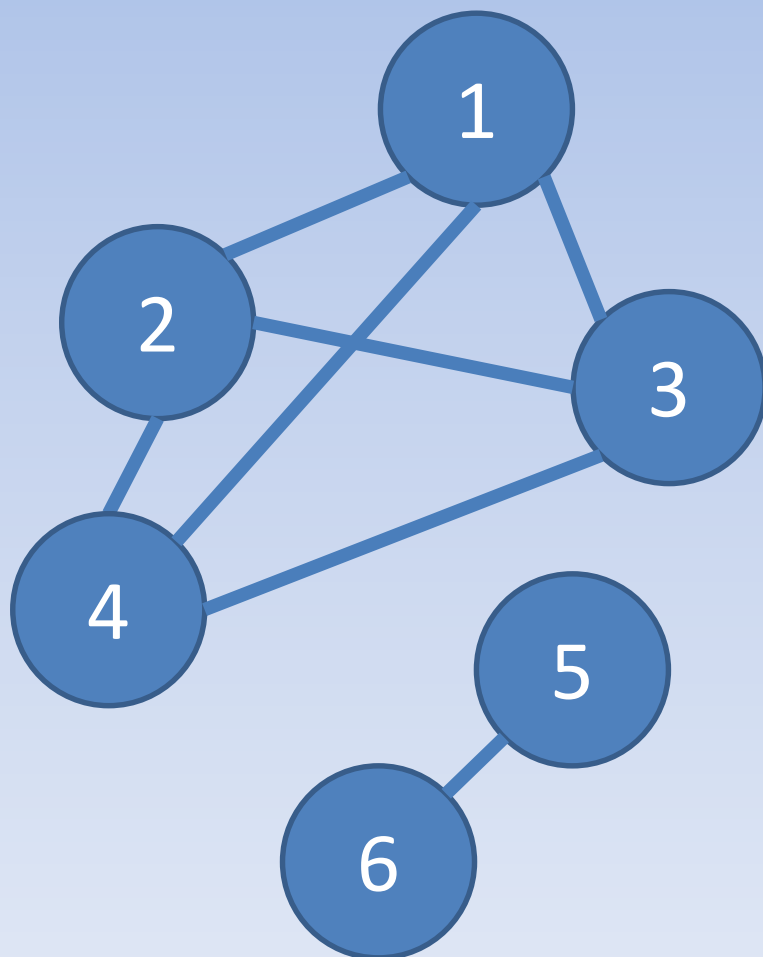
4 1

4 3

2 3

6 5

-1

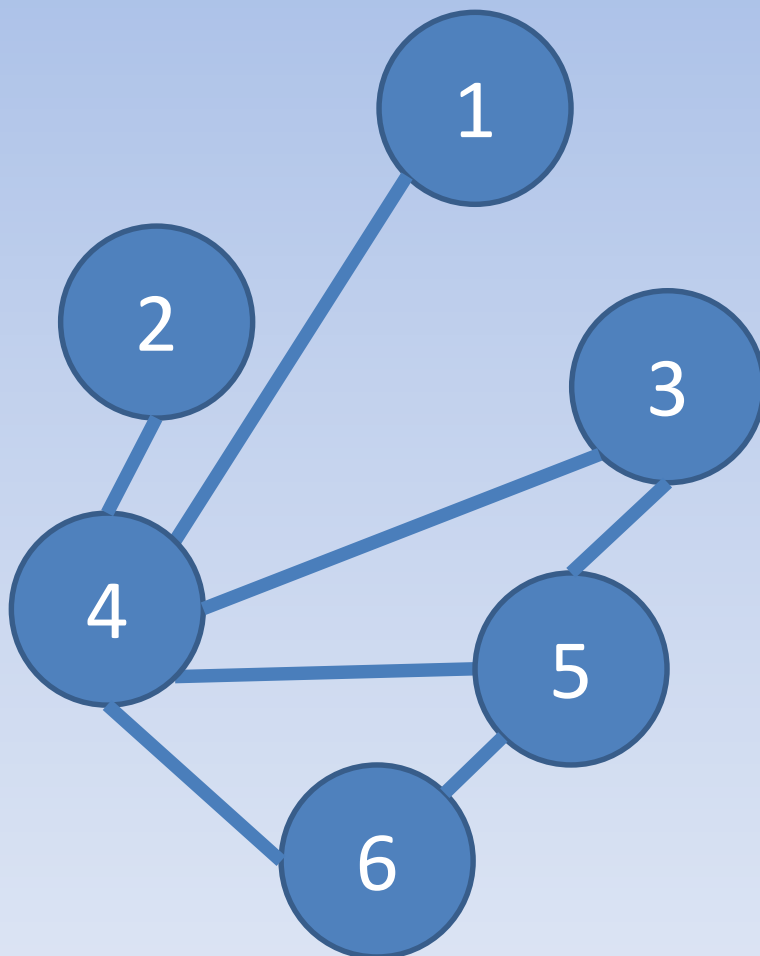


Sample output 2

n

Sample input 3

6
4 3
4 5
4 6
1 4
2 4
3 5
5 6
-1

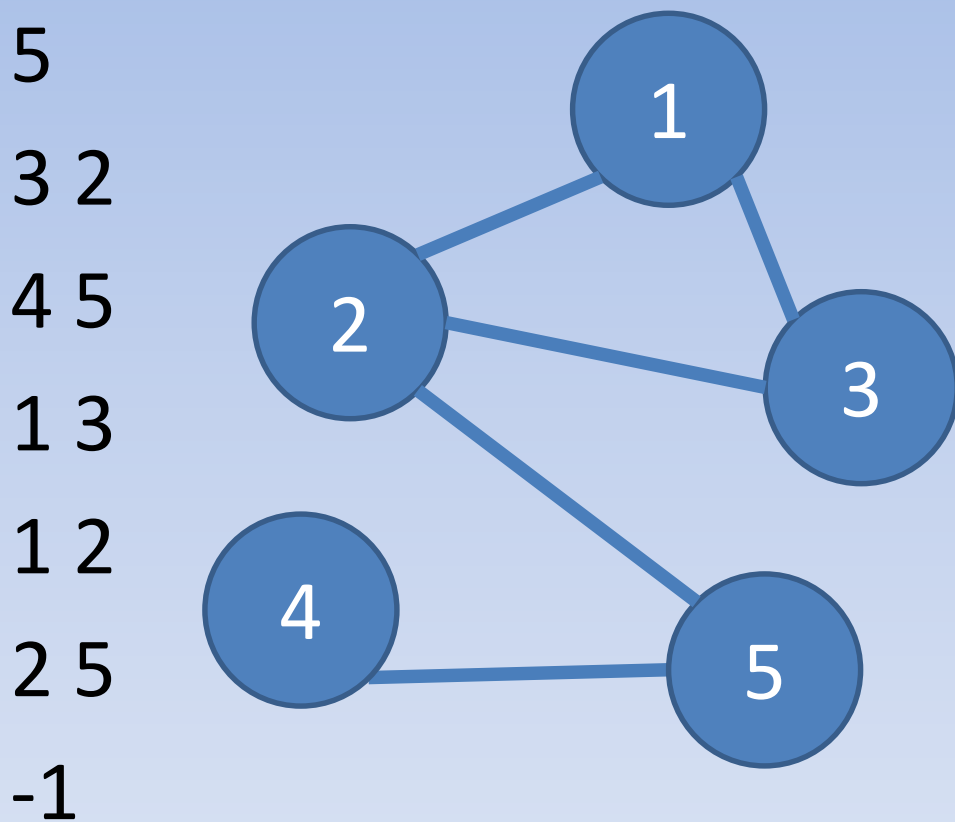


Sample output 3

y

There are at least three cycles.

Sample input 4

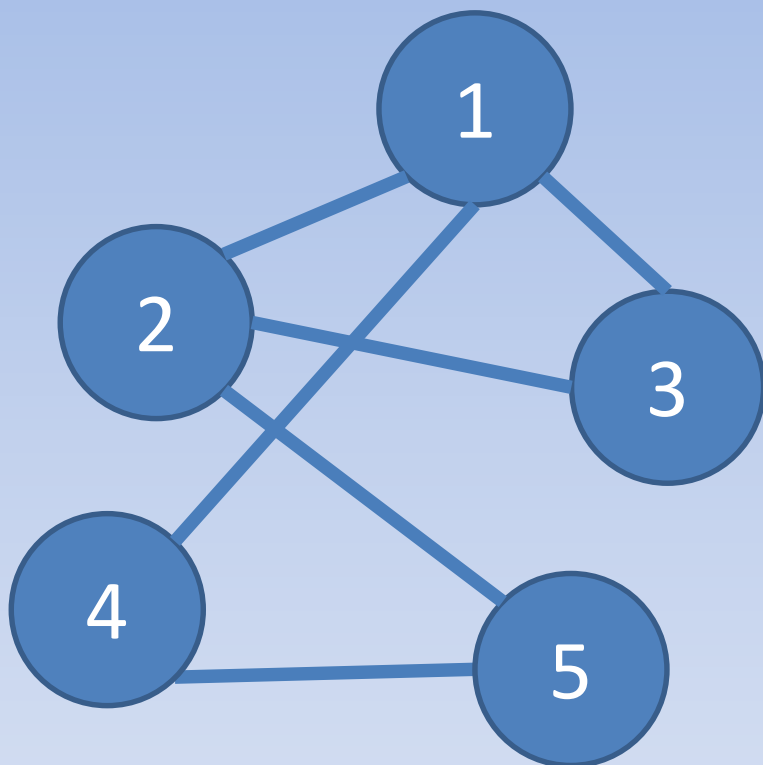


Sample output 4

n

Sample input 5

5
3 2
4 5
1 3
1 2
2 5
4 1
-1

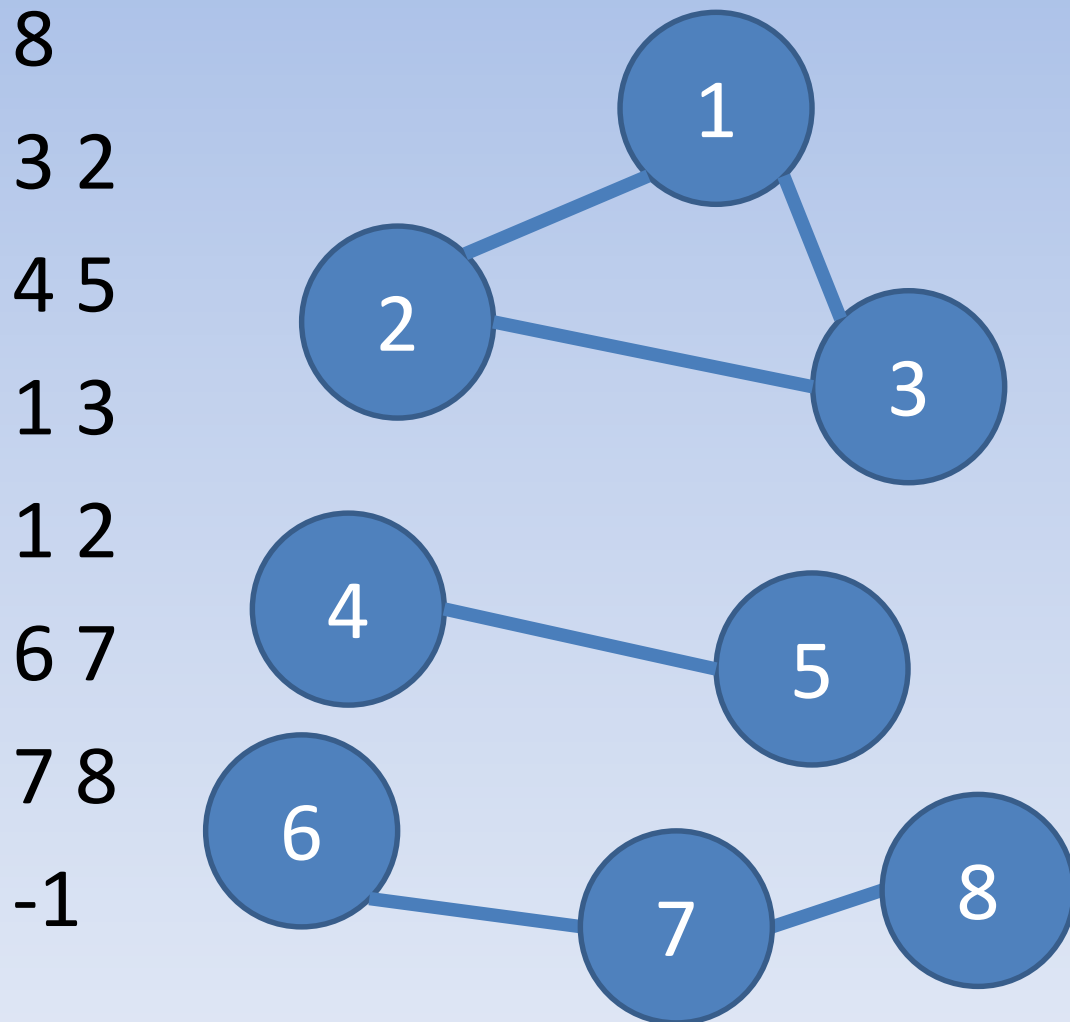


Sample output 5

y

There are at least three cycles.

Sample input 6

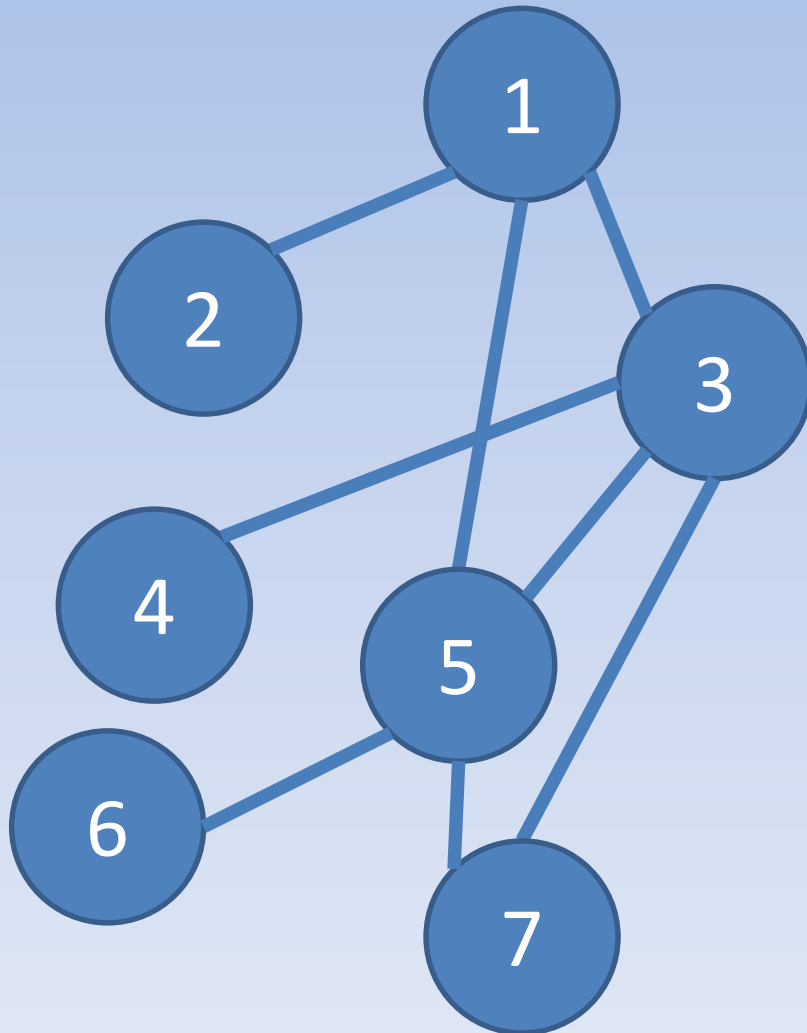


Sample output 6

n

Sample input 7

7
2 1
1 3
1 5
3 7
3 5
3 4
6 5
5 7
-1



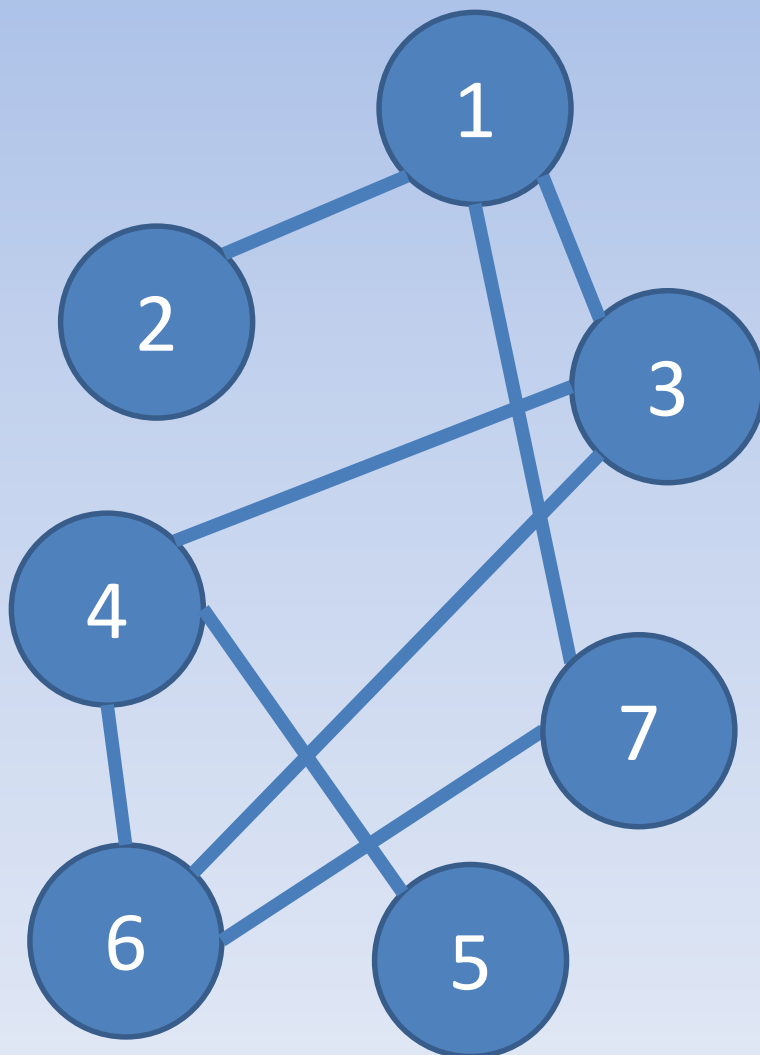
Sample output 7

y

There are at least three cycles.

Sample input 8

7
1 2
1 3
3 4
4 5
6 4
1 7
3 6
7 6
-1

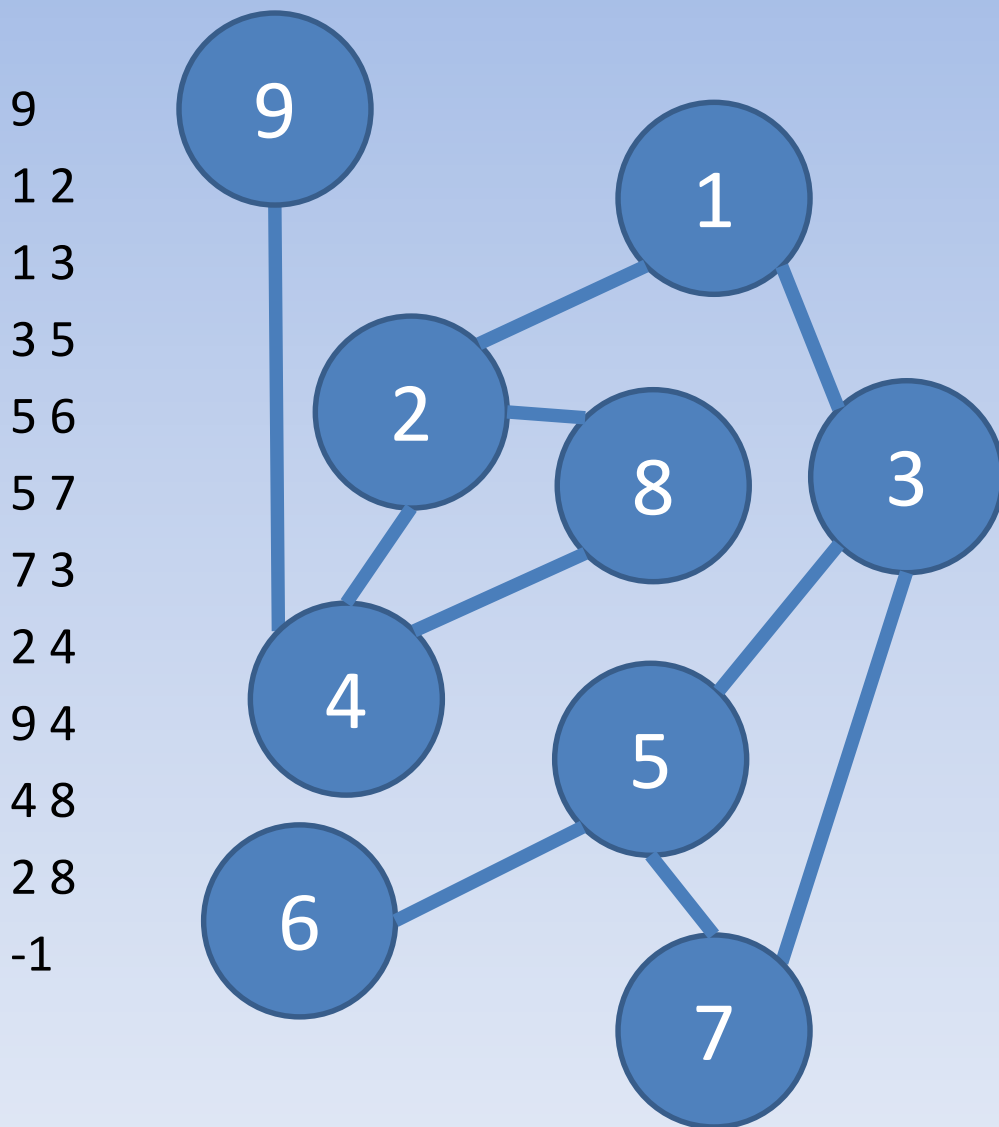


Sample output 8

y

There are at least three cycles.

Sample input 9



9

1 2

1 3

3 5

5 6

5 7

7 3

2 4

9 4

4 8

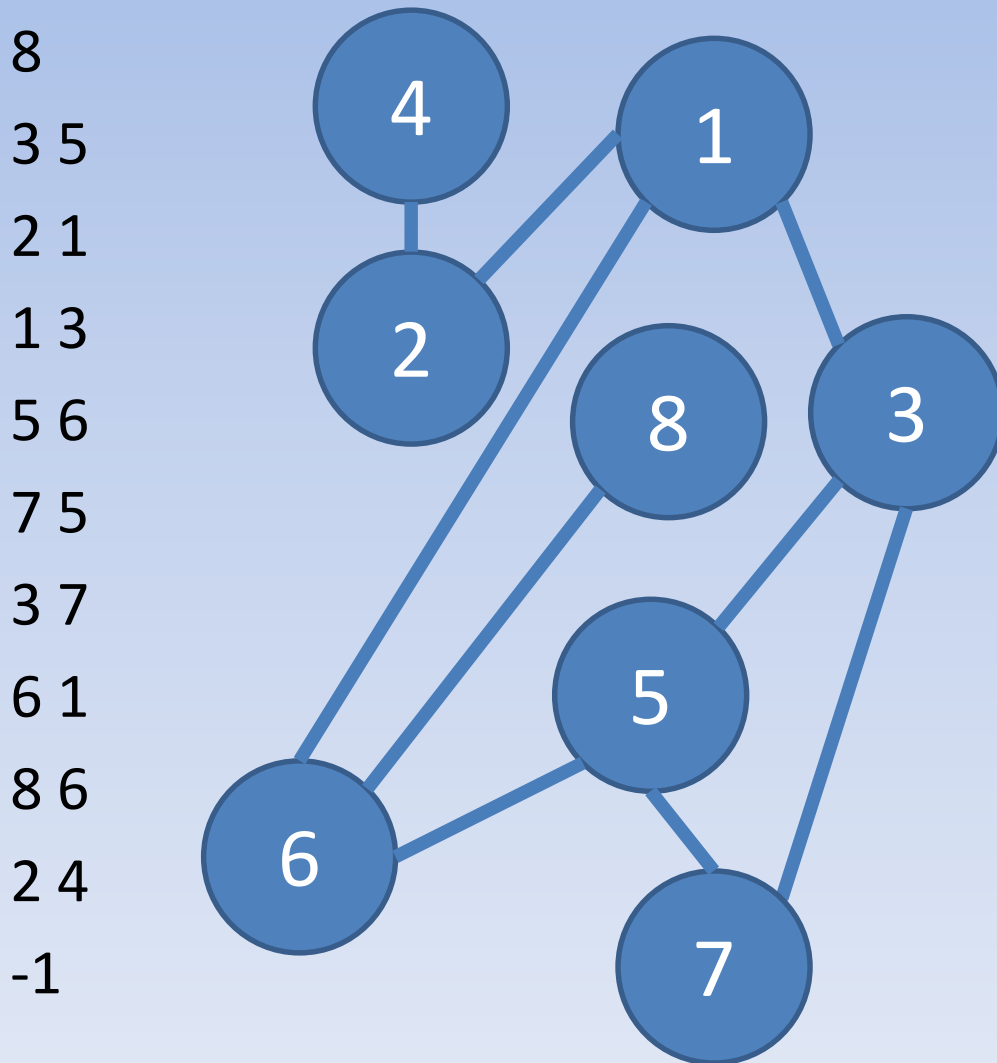
2 8

-1

Sample output 9

y

Sample input 10

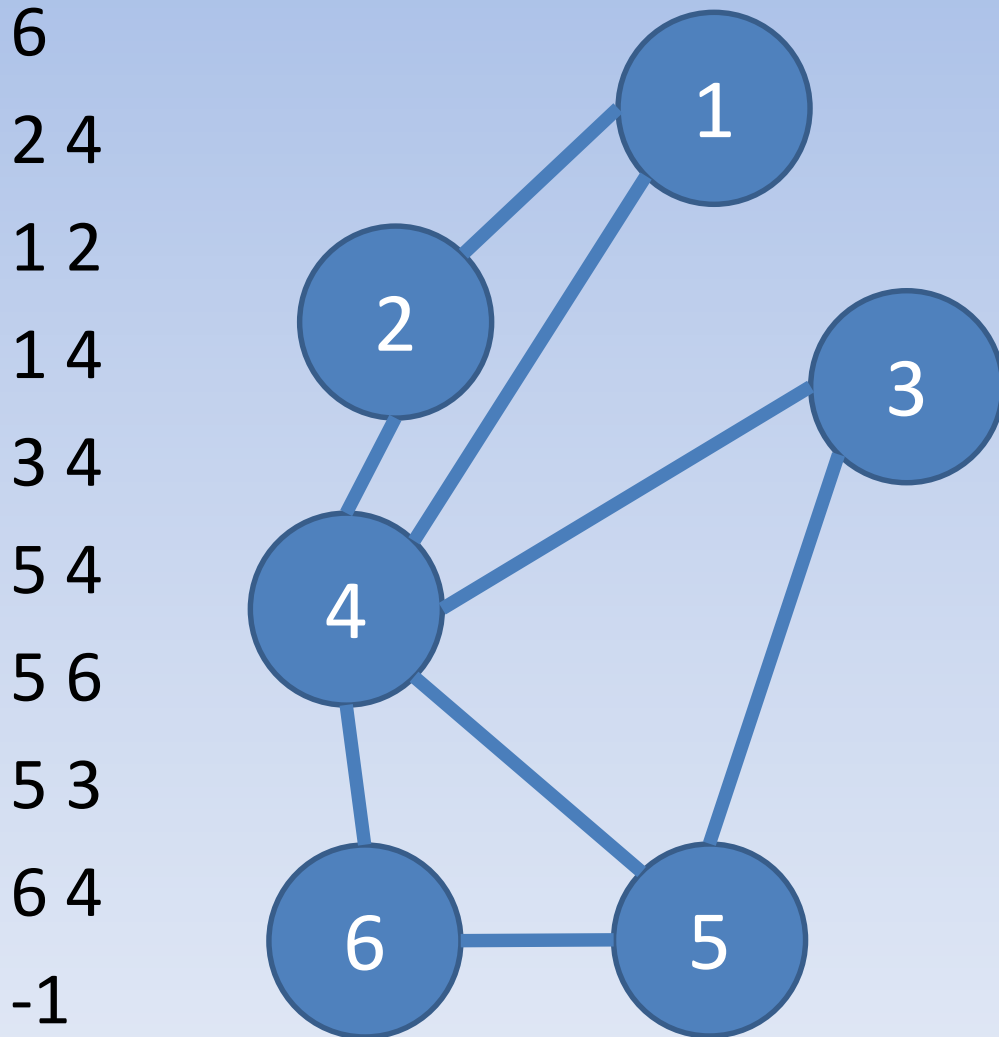


Sample output 10

y

There are at least three cycles.

Sample input 11



6
2 4
1 2
1 4
3 4
5 4
5 6
5 3
6 4
-1

Sample output 11

y

There are at least three cycles.

格式

- 因為我會用**diff**直接對答案，所以麻煩依照格式輸出，謝謝大家

Deadline

- TBA

判斷方法

- 兩條不在spanning tree上的邊會各自與spanning tree的一些邊形成一個cycle，如果這兩個cycles有共用邊，則全圖有三個cycles，否則全圖恰有兩個cycles

Comments?

